Topic 14 — Environmental Impact Assessment

Covers syllabus (click to expand)

Environmental Impact Assessment (EIA) (पर्यावरण प्रभाव आकलन) | Strategic Environmental Assessment (SEA) (रणनीतिक पर्यावरण आकलन)

Sources baked in: NCERT Geography/Biology, EPA 1986, EIA Notification 2006, NGT Act 2010, UNECE SEA Protocol 2003, PARIVESH, UPPCS/UPSC PYQs **Weight:** ★★ Medium–High — Category A/B traps, four stages, MoEFCC vs SEIAA vs CPCB, EIA vs SEA **Last verified:** August 2026

Current Affairs (this topic)

- EIA notification amendments and Category A/B traps are the living CA layer.
- Keep screening–scoping–appraisal–monitoring order stable.

14.1 Environmental Impact Assessment (EIA)

Identity: EPA 1986 | EIA Notification 2006 | Category A/B | Four stages | EC before construction

- **Environmental Impact Assessment (EIA)** is a systematic study of a proposed project's likely environmental, social, and health effects **before** major decisions are taken.
- The goal is **anticipatory prevention** — reject, redesign, or condition projects rather than repair damage after construction.
- The **Environment (Protection) Act, 1986** is the legal base. **Section 3** empowers the Centre to notify rules requiring prior clearance.
- The **first EIA Notification** came in **1994**. The current framework is the **EIA Notification 2006** dated **14 September 2006**.
- **Environmental Clearance (EC)** is mandatory for **listed projects** in the Schedule before construction or land preparation.
- **Category A** projects are large, high-impact projects appraised centrally by **MoEFCC** through the **Expert Appraisal Committee (EAC)**.
- **Category B** projects are appraised by the **State Environment Impact Assessment Authority (SEIAA)** through the **State Expert Appraisal Committee (SEAC)**.
- **Screening** applies to **Category B only** and splits projects into **B1** (full EIA required) or **B2** (no full EIA report).
- Under the **General Condition**, a Category B project in a **sensitive area** — national park, sanctuary, ESZ, CRZ, inter-state boundary, or critically polluted zone — is treated as **Category A**.
- The **four-stage process** runs: **(1) Screening**, **(2) Scoping** (ToR fixed), **(3) Public Consultation** (public hearing + written submissions), **(4) Appraisal** (EAC/SEAC recommendation).

- An **EIA report** covers baseline data, impact prediction, mitigation measures, and an **Environmental Management Plan (EMP)**.
- **Rapid EIA** uses one season of baseline data. **Comprehensive EIA** covers all seasons and is more reliable for monsoon-dry variation.
- **CPCB and SPCBs** set pollution standards and monitor compliance. They **do not grant EC**.
- **PARIVESH** is MoEFCC's online single-window portal for EC applications and tracking.
- **NGT (National Green Tribunal)**, under the **NGT Act 2010**, hears appeals against EC decisions.
- EC must be obtained **before** construction. **Ex-post facto clearance** after work has begun is illegal and controversial.

> **Logic:** Carrying capacity — waste generation must stay within the environment's absorption limit; that is what EIA evaluates.
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> **Logic:** Org trap — **CGWB** reports groundwater quality; **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) does not grant EC. EC = **MoEFCC/SEIAA**.

Category	Authority	Screening?	Full EIA?
A	MoEFCC + EAC	No	Usually yes
B1	SEIAA + SEAC	Yes	Yes
B2	SEIAA/DEIAA	Yes	No full report
B + General Condition	MoEFCC (elevated)	—	As Category A
Stage	What happens		
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Screening	B → B1 or B2		
Scoping	ToR for EIA study		
Public Consultation	Public hearing + comments		
Appraisal	Recommend grant/reject EC		

14.2 Strategic Environmental Assessment (SEA)

Identity: PPP level | Upstream of EIA | No standalone Indian law | SEA Protocol 2003

- **Strategic Environmental Assessment (SEA)** evaluates the environmental consequences of **policies, plans, and programmes (PPP)** at the earliest planning stage.
- **SEA sits upstream of EIA**. It asks whether a development path is wise **before** individual projects are designed.
- **EIA** can only accept, modify, or reject **one pre-selected project**. It cannot question whether an entire sector plan — such as many dams in one basin — is environmentally sound.

- **SEA assesses cumulative impacts** when multiple projects together may overwhelm an ecosystem (पारिस्थितिकी तंत्र) even if each passes individual EIA.
- **Public participation** in SEA happens when policy choices are still flexible — earlier and cheaper than late-stage project hearings.
- **Rio Declaration (1992) Principle 17** calls for **EIA** for proposed activities with significant adverse impact. It does **not** mandate SEA by name.
- **Espoo Convention (1991)** covers **transboundary EIA**. The **SEA Protocol (2003)** signed at **Kiev** extends assessment to plans and programmes.
- The **EU SEA Directive (2001/42/EC)** makes SEA mandatory for listed public plans in member states.
- **India has robust project-level EIA** under the 2006 Notification but **no standalone SEA legislation** as of current law.
- Partial strategic tools in India include **CRZ notifications**, **Eco-Sensitive Zone (ESZ) (पर्यावरण-संवेदनशील क्षेत्र) notifications**, and basin-level planning — but these are not a full SEA law.

> **Logic:** Statement "India has mandatory SEA law equivalent to EIA Notification" = **false**.

Feature	EIA	SEA
Level	Individual project	Policy/plan/programme
Timing	After project concept	Before projects chosen
India law	Mandatory (2006)	No standalone law
Cumulative impacts	Limited	Core focus
Example	EC for one thermal plant	National sector plan before siting
Instrument	Year	Scope
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Rio Principle 17	1992	Project EIA norm
Espoo Convention	1991	Transboundary EIA
SEA Protocol (Kiev)	2003	Plans/programmes
EU SEA Directive	2001/42/EC	EU mandatory SEA
India EIA Notification	2006	Project EC only